Ultrasound-Guided Robotic Autonomous Operation Based on Real-Time Deformation Tracking and Prediction

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***Abstract*—Robotic ultrasound (US)-guided surgery is one of the most common clinical workflows for many diseases. Intraoperative tissue deformation is a key challenge faced by robots in realizing autonomous surgery. This study develops a US-guided robotic autonomous operation scheme based on deformation tracking and prediction, which can overcome the adverse effects of intraoperative deformation. First, a deformation tracking method based on real-time nonrigid image registration is proposed to track tissue and path deformation in real time. Simultaneously, we propose a deformation prediction model that is updated by the forgetting factor recursive least squares for predicting the next deformation according to historical tracked data. Then, a time-varying vector field navigation law is incorporated to guide the robot following the dynamically deformed surgical path, therefore compensating for deformation. To verify the performance of the proposed method, a dual-arm robotic US surgical navigation system prototype is constructed, and *in vitro* experiments in a deformable environment are conducted. The results demonstrated that our method can guide the robot to compensate for tissue deformation in real-time with a dynamic path following error within 1 mm.**

***Index Terms*—Autonomous robotic navigation, deformation prediction, dynamic path following, motion tracking, ultrasound (US)-guided surgery.**

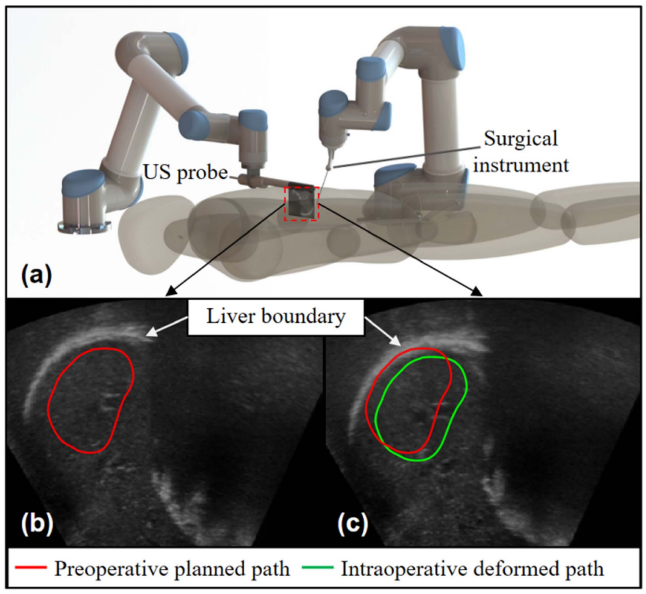


Fig. 1. (a) Illustration of robotic US-guided surgery. (b) Preoperative liver US image and planned path. (c) When deformation occurs, the initial path becomes invalid and even causes danger due to the close to the liver border.

I. Introduction

ROBOTIC ultrasound (US)-guided surgery has become one of the most commonly used clinical workflows currently [1] due to the accessibility, low cost, nonionizing nature, and real-time capability of the US. These workflows are widely distributed in fields such as robotic imaging scanning [2], interventions of various cancers [3], and US-guided robotic operations [4]. This paradigm of combining US imaging with robotics simplified clinical procedures, improved surgical precision, and obtained several successful commercial applications [5]. An important trend in the robotic US-guided surgery is robotic autonomous surgery which is attracting more and more attention [6], [7]. Compared to current manual or teleoperated robotic surgeries, it has the potential to substantially improve efficiency, safety, and consistency [8], thus reducing the surgeon's labor time. A major challenge of robotic autonomous surgery is dynamic intraoperative soft tissue deformation affected by respiration, heartbeat, or movement of surgical instruments, etc. This deformation may lead to the failure of preoperative planning (see Fig. 1), which brings higher requirements to robots for real-time motion tracking and autonomous navigation capabilities in a deformable environment [9].

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As one of the main clinical real-time imaging methods, robotic US-guided surgery has attracted much attention in the past decade. In 2012, Hungr et al. proposed a US robotic prostate brachytherapy system [3]. Through a 3-D US image registration algorithm, this system can track intraoperative prostate motion. However, the time required for a single registration was about 6–8 s. In 2016, Selmi et al. [10] proposed a feature-based registration method for robotic prostate biopsy, which took approximately 2 s for each motion tracking. In 2022, Jiang et al. [11] proposed an autonomous vascular screening robot based on real-time ultrasound image feedback, which optimized the robot pose in real-time based on vascular structure reconstruction using a deep learning model U-Net. Recently, Tan et al. [12] developed a real-time autonomous US robot trajectory planning framework to achieve global scanning of breast tissue and real-time local detection of suspicious tumor targets, it can make the US image of suspicious tumor targets be displayed stably in the center of the acoustic window. US-guided robotic navigation methods were also widely applied in microrobots [13]. US imaging was usually used to provide feedback to robotic systems and control the microrobot's motion with active steering [14]. The above methods have significantly promoted the development of US-guided robotic operation technology. However, these methods either did not consider the tissue deformation or were limited by the real-time capability of tracking and could not quickly guide the robot to compensate for the deformation.

To deal with the problem of intraoperative real-time deformation compensation, a promising solution is to integrate visual perception into the robotic workflow. Intraoperative imaging methods such as US or endoscopic cameras were applied for detecting and tracking target tissues as feedback to the robot [15]. Wang et al. [16] proposed an autonomous robot for incision stapling. Based on binocular vision, they applied the Canny algorithm to conduct real-time detection of the incision's position. Kam et al. [17] proposed an autonomous robotic system for vaginal cuff closure, they used a deep learning method for autonomous markerless soft tissue tracking and developed a model-based suture planning method. These vision-guided methods showed considerable potential to improve autonomy in a lot of surgical tasks [18]. US-guided surgeries often occur in environments with narrow fields of view, such as the urethra or bladder, so vision-based methods are limited, but some promising deformation tracking methods also appeared. In 2018, Samei et al. [19] developed a real-time registration method for US-guided laparoscopic radical prostatectomy based on a finite-element model. They realized intraoperative nonrigid motion tracking of 264 ms. Based on this method, they then conducted preliminary clinical validation of Da Vinci in robot-assisted radical prostatectomy in 2020 [20]; however, the tracking results were not used as feedback for the robot. In 2022, Tu et al. [21] proposed a US-guided and mixed reality based robotic cervical pedicle screw placement system, a shape prior model based on principal component analysis was applied for tissue deformation computing at a speed of 247 ms to provide intraoperative guidance. However, currently, studies focusing on intraoperative robotic automatic deformation compensation technology are still lacking in robotic US-guided surgeries.

This study proposed a US-guided robotic autonomous operation method in an intraoperative deformable environment. First, based on real-time registration of US images, it could track intraoperative tissue deformation, and simultaneously predict the next deformation based on historical registered data. Subsequently, by introducing a time-varying navigation law with the remote center of motion (RCM) constraint, it guided the robot to compensate for tissue deformation in real time. To the best of the authors' knowledge, this article is the first study that can track and predict tissue deformation during robotic US-guided surgery while guiding the robot to compensate for the deformation in real time.

The main contributions of this study are as follows.

1) A path deformation tracking method based on real-time nonrigid US image registration is proposed, through accelerating the registration process, it can track tissue and path deformation in real-time ($\leq$ 30 ms).
2) A path deformation prediction model is proposed, which can adaptively update parameters based on forgetting factor recursive least squares (FFRLS), predicting the next path deformation based on historical registration data.
3) A US-guided robotic autonomous operation scheme is developed by introducing a time-varying vector field with RCM constraint for feedforward following the dynamic path, compensating for tissue deformation in real time.
4) A dual-arm robotic US surgical navigation system prototype was constructed. *In vitro* experiments demonstrated that the proposed method can compensate for tissue deformation in real time with a dynamic path following error $\leq$ 1 mm.

II. Problem Formulation

The proposed US-guided robotic operation system in this article is illustrated in Fig. 2(a). This system consists of two robots: robot A holds a US probe for obtaining real-time US image streaming, whereas robot B carries a surgical instrument and performs related surgical operations represented by robotic paths on designated areas in the US image. In this study, pixels in the US image are considered as a point set with intensities and spatial positions. The coordinate relationship between the pixel positions ${}^{\mathrm{US}}\mathbf{p}$ in the US image and the base of robot B is ${}^{B}\mathbf{p} = {}^{B}_{\mathrm{US}}\mathbf{T}\,{}^{\mathrm{US}}\mathbf{p}$, where $\mathbf{T}$ is a rigid transformation. In the remainder of this article, unless otherwise specified, coordinate references are based on the coordinates of the US image.

Assumed that the preoperative planned surgical path in the US image space is a twice differentiable curve $\mathcal{P}_0 = \mathbf{r}(s, t_0)$, where $\mathbf{r}(s, t_0)$ is the parameterization for $\mathcal{P}_0$ with s the arc-length parameter. Let $\mathbf{x}$ and $\dot{\mathbf{x}}$ be the position and velocity of the tip of the surgical instrument of robot B (abbreviated as robot tip), without tissue deformation, a navigation law $\dot{\mathbf{x}} = \Psi(\mathbf{x}, t)$ can be designed to make the tip of robot B's trajectory converge to and traverse the preplanned path $\mathcal{P}_0$.

The most intuitive manifestation of intraoperative tissue deformation is the deformation of US images. Therefore, in this

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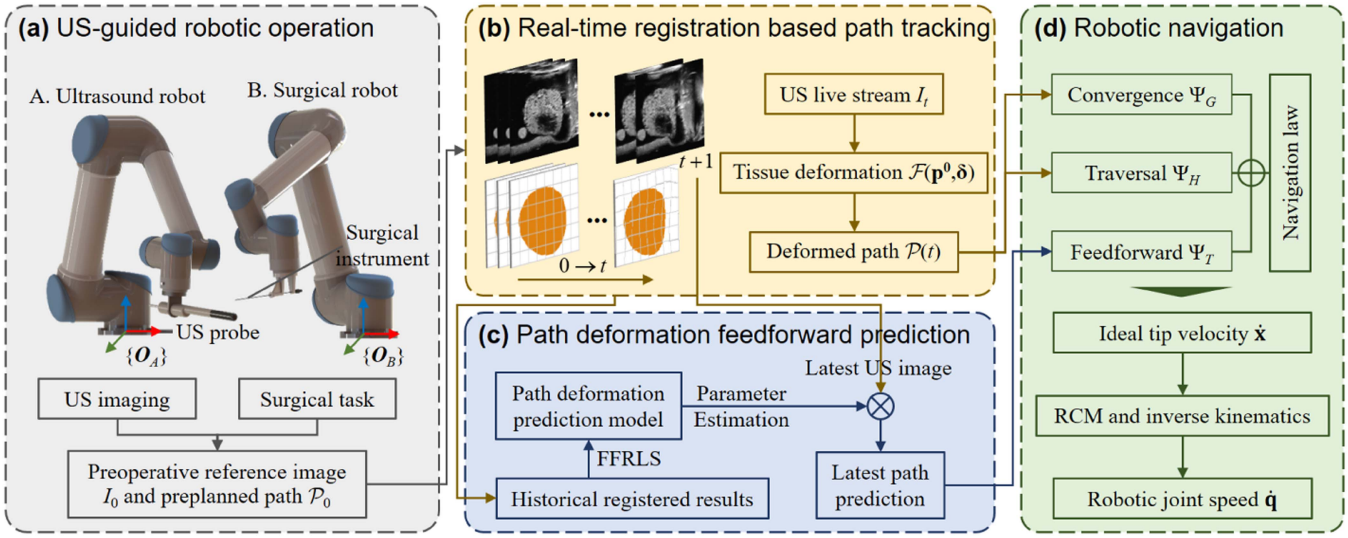


Fig. 2. Workflow of the US-guided robotic autonomous operation scheme proposed in this article. (a) US-guided robotic rotation. (b) Real-time registration-based path tracking. (c) Robotic navigation.

study, we equate the two concepts of tissue deformation and US image deformation. The deformation of US image is described by the deformation field [22]. The deformation field $\mathbf{u}$ describes the displacement of each pixel in the US image space. Let $\mathbf{p}^0 = [p_x^0, p_y^0]^{\mathrm{T}}$ represents the position of a certain pixel in the image, then $\mathbf{u} = [u_x, u_y]^{\mathrm{T}}$ represents the displacement of $\mathbf{p}^0$. Due to the nonuniformity of human tissue, image deformation is usually nonrigid, which requires a nonrigid registration method to solve the deformation field for motion tracking. Afterward, the real-time intraoperative deformed path $\mathcal{P}(t)$ can be calculated based on the current deformation field of the US image. However, it is quite difficult to solve a nonrigid deformation field of US images in real time.

Therefore, the problem addressed in this study is as follows.

In robotic US-guided surgeries, given the preoperative surgical path $\mathcal{P}_0$, during the surgery, the system can calculate the deformation of tissue and path $\mathcal{P}(t)$ in real time based on the US image stream while guiding the trajectories of the robot tip $\dot{\mathbf{x}} = \Psi(\mathbf{x}, t)$ to follow the dynamic surgical path $\mathcal{P}(t)$ to compensate for tissue deformation.

III. Methodology

The workflow of the robotic US navigation scheme proposed in this article is shown in Fig. 2. First, the deformation of tissue and path is derived based on real-time registration. Then, a feedforward prediction model is proposed to predict the next deformation. Finally, a navigation law was incorporated to make the robot follow the dynamically deformed path.

A. Real-Time Registration Based Deformation Tracking

The deformation tracking is realized by image registration. Typically, it consists of three main parts: deformation field, loss function, and optimization. The deformation field adopted in this article is the 2-D form of free-form deformation (FFD) [23].

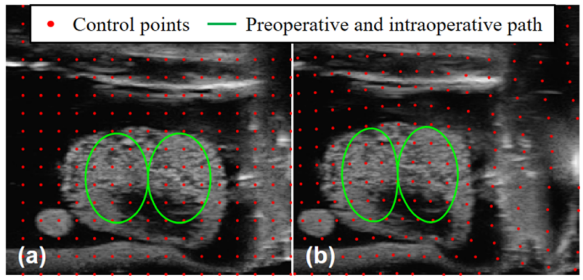


Fig. 3. (a) Preoperative US reference image, initial control points, and preplanned surgical path. (b) When the positions of the control point are moved, the US reference image and surgical path will be deformed accordingly.

Given the preoperative reference US image I_0 and corresponding real-time image $I_t : t \in \{1, 2, \cdots, T\}$ with the size of $M \times M$, the position of any pixel in the reference image is represented by $\mathbf{p}^0 = [p_x^0, p_y^0]^{\mathrm{T}}$, where p_x^0 and p_y^0 are the coordinates of $\mathbf{p}^0$ in the x and y directions. The task of registration is to optimize the displacement $\mathbf{u}$ of every pixel so that the deformed (registered) reference image I_0^D and the real-time image I_t can be as similar as possible.

FFD deforms the US image by manipulating the movement of control points. As shown in Fig. 3(a), control points $\boldsymbol{\lambda} = [\boldsymbol{\lambda}_x, \boldsymbol{\lambda}_y]^{\mathrm{T}}$ are sparse, regular grid point coordinates defined in image space with the size of $K \times K$ and pixel spacing constant of $\Delta\rho$. Also, we define the displacement of control points as $\boldsymbol{\delta} = [\boldsymbol{\delta}_x, \boldsymbol{\delta}_y]^{\mathrm{T}}$. According to the FFD model, given the positions λ and displacement δ of the control points, the deformation $\mathbf{u}$ of pixel $\mathbf{p}^0$ in the US image is calculated as follows:

$$\mathbf{u} = \mathcal{F}(\mathbf{p}^0, \boldsymbol{\delta}) = \sum_{\mathbf{j}} \boldsymbol{\delta}_{\mathbf{j}} \Phi^{(3)}((\mathbf{p}^0 - \boldsymbol{\lambda}_{\mathbf{j}})/\Delta\rho) \tag{1}$$

where $\mathbf{j}$ is the index of the control points. $\Phi^{(3)}(\mathbf{v}) = \beta^{(3)}(x)\beta^{(3)}(y)$ is a convolution kernel, and $\mathbf{v} = [x, y]^{\mathrm{T}}$ is a

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spatial position vector. The convolution kernel Φ smoothly interpolates the control points' displacement into nearby pixels, it can be represented as a cubic B-spline in the x-direction

$$\beta^{(3)}(x)=\begin{cases}(4-6x^2+3|x|^3)/6, & 0\le|x|<1\\ (2-|x|)^3/6, & 1\le|x|<2\\ 0, & 2\le|x|\end{cases}. \quad (2)$$

$\beta^{(3)}(y)$ can be calculated in the same way. As shown in Fig. 3, when the positions of the control points are moved from (a) to (b), the US image will also be deformed accordingly. The similarity between the deformed US reference image and the real-time image I_t is described by a loss function. It is the sum of a similarity function term and a smooth regularization term [22] with γ as a regularization constant

$$C_T(I_0,I_t,\boldsymbol{\delta})=C_D(I_0,I_t,\boldsymbol{\delta})+\gamma\cdot C_R(\mathcal{F}(\mathbf{p}^0,\boldsymbol{\delta})) \quad (3)$$

where the first similarity function term C_D is the sum of the squared difference (SSD) between two images

$$C_D(I_0,I_t,\boldsymbol{\delta})=\frac{1}{2}\left\|I_0^D(\mathbf{p})-I_t\right\|_2^2=\frac{1}{2}\left\|I_0^D(\mathbf{p}^0,\mathcal{F}(\mathbf{p}^0,\boldsymbol{\delta}))-I_t\right\|_2^2 \quad (4)$$

where $\mathbf{p}=\mathbf{p}^0+\mathbf{u}$ is the deformed position of pixel $\mathbf{p}^0$. Based on the above analysis, the deformable registration between images can be summarized as an optimization problem: during the deformation of US image, the displacement δ of the control points is optimized to find an optimal displacement field $\mathbf{u}$ between the reference image I_0 and moving image I_t at time t

$$\boldsymbol{\delta}_t^*=\arg\min_{\boldsymbol{\delta}_t}C_D(I_0,I_t,\boldsymbol{\delta}_t)+\gamma\cdot C_R(\mathcal{F}(\mathbf{p}^0,\boldsymbol{\delta}_t)). \quad (5)$$

Since the US images are acquired through a video stream, the reference image is registered with every sequence to solve the real-time deformation field. When registering a new sequence I_t at time t, the previously registered results $\boldsymbol{\delta}_{t-1}^*$ at time t-1 can be regarded as prior. Hence, the continuous registration problem from time t-1 to time t is expressed as

$$\boldsymbol{\delta}_{t-1,t}^*=\arg\min_{\boldsymbol{\delta}_{t-1,t}}C_D(I_0,I_t,\boldsymbol{\delta}_{t-1}^*+\boldsymbol{\delta}_{t-1,t})+\gamma\cdot C_R(\mathcal{F}(\mathbf{p}^0,\boldsymbol{\delta}_{t-1}^*+\boldsymbol{\delta}_{t-1,t})). \quad (6)$$

We define the displacement of the control points between the two sequences from time t-1 to t as the current instantaneous speed of the control points: $\boldsymbol{\upsilon}_t=\boldsymbol{\delta}_{t-1,t}^*$. Thus, the control points' displacement at time t can also be written as $\boldsymbol{\delta}_t^*=\boldsymbol{\delta}_{t-1}^*+\boldsymbol{\upsilon}_t$.

After registration, the deformed path can be obtained based on the optimized control points' displacement (see Fig. 3)

$$\mathcal{P}(t)=\mathcal{P}_0+\mathcal{F}(\mathbf{r},\boldsymbol{\delta}_t^*)=\mathcal{P}_0+\sum_{\mathbf{j}}\boldsymbol{\delta}_t^*\cdot\Phi^{(3)}((\mathbf{r}-\boldsymbol{\lambda}_\mathbf{j})/\Delta\rho). \quad (7)$$

Generally, solving the nonrigid registration problem (6) requires simultaneously optimizing hundreds of variables, which is difficult to complete in real time. In this study, we accelerate the nonrigid registration through analytic gradients [19], cascade convolution [24], and trust-region optimization, which enables the registration can be completed within 30 ms for the US images with the size of 256 × 256.

B. Path Deformation Feedforward Prediction

In the previous section, by accelerating nonrigid registration (6), the speed of deformation tracing can be increased to within 30 ms. However, this means that each calculated path deformation is the result of at least 30 ms ago. If other factors are taken into consideration, this delay will become even larger. Hence, to further improve the tracking accuracy, a feedforward deformation prediction model is proposed in this section.

From (6), it can be found that the displacement of the control points is a function of the SSD loss C_D and the regularization C_R, where the SSD function term makes I_0 and I_t as similar as possible, and the regularization term maintains the smoothness of pixels' deformation, similar to the "external force" and "internal force" of soft tissue deformation.

$$\boldsymbol{\upsilon}_t=\boldsymbol{\delta}_{t-1,t}^*=f(C_D)+f(C_R). \quad (8)$$

Based on the study of Le Folgoc et al. [25], the SSD function and pixels' displacement are related as quadratic terms. Let the symbol vec(·) be an operator that expands a scalar into a column vector, and the symbol diag(·) generate a diagonal matrix with a scalar as the main diagonal. A deformation feedforward prediction model is proposed here

$$\boldsymbol{\upsilon}_t=\text{diag}\left(\left(C_{D|t-1,t}\right)^{1/2}\right)\cdot\boldsymbol{\alpha}+\text{vec}(C_{R|t-1,t})\cdot\beta \quad (9)$$

where α and β are the response coefficients of the control points displacement relative to C_D and C_R, respectively. The dimensions of α, vec(·), and diag(·) operator are the same as the control point velocity υ. Through (9), the relationship between the displacement of every control point and the loss functions can be established. At the same time, by integrating (9), we derive the following regression problem:

$$\mathbf{y}_t=\boldsymbol{\psi}_t^{\mathrm{T}}\cdot\boldsymbol{\theta}_t+\bar{\varepsilon}_t. \quad (10)$$

Among this, $\boldsymbol{\psi}_t^{\mathrm{T}}$ is the regression matrix, $\boldsymbol{\theta}_t$ is the vector of the parameters, and $\bar{\varepsilon}_t$ represents the modeling error and other noise. Each item is expressed as

$$\mathbf{y}_t=\boldsymbol{\upsilon}_t,\boldsymbol{\psi}_t^{\mathrm{T}}=[\text{diag}((C_{D|t-1,t})^{1/2})|\text{vec}(C_{R|t-1,t})]\boldsymbol{\theta}_t=[\boldsymbol{\alpha}^{\mathrm{T}},\beta]^{\mathrm{T}}. \quad (11)$$

It is noted that the regression parameter θ_t is not constant, it varies with the speed and direction of tissue deformation. To adapt to this dynamic variation, the forgetting factor recursive least squares (FFRLS) method is applied to fit parameters dynamically [26]. FFRLS has a forgetting factor to estimate variable parameters, its updated equations can be written as

$$\mathbf{K}_{t+1}=\mathbf{P}_t\cdot\boldsymbol{\psi}_{t+1}\cdot(\mu\mathbf{I}+\boldsymbol{\psi}_{t+1}^{\mathrm{T}}\cdot\mathbf{P}_t\cdot\boldsymbol{\psi}_{t+1})^{-1} \quad (12)$$

$$\hat{\theta}_{t+1}=\hat{\theta}_t+\mathbf{K}_{t+1}\cdot(\mathbf{y}_{t+1}-\boldsymbol{\psi}_{t+1}^{\mathrm{T}}\cdot\hat{\theta}_t) \quad (13)$$

$$\mathbf{P}_{t+1}=\frac{1}{\mu}\cdot(\mathbf{I}-\mathbf{K}_{t+1}\cdot\boldsymbol{\psi}_{t+1}^{\mathrm{T}})\cdot\mathbf{P}_t \quad (14)$$

where $\mathbf{P}=(\boldsymbol{\psi}_t\cdot\boldsymbol{\psi}_t^{\mathrm{T}})^{-1}$ is the covariance matrix, $\mathbf{K}$ is the gain value representing the update effect of new data, $\mathbf{I}$ is an

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identity matrix, and $\mu \in (0, 1]$ is the forgetting factor. A smaller forgetting factor allows for faster adaptation to new data, but may also make predictions unstable.

Based on the regression of the historical registration data, when a new US sequence I_{t+1} is captured, only C_D and C_R need to be calculated to predict the latest tissue deformation

$$\hat{v}_{t+1} = \boldsymbol{\psi}_{t+1}^{\mathrm{T}} \cdot \boldsymbol{\theta}_t. \tag{15}$$

Subsequently, the prediction of path deformation for the time step $t+1$ can be obtained

$$\hat{\mathcal{P}}(t+1) = \hat{\mathbf{r}}(s, t+1) = \boldsymbol{\mathcal{P}}_0 + \sum_{\mathbf{j}} (\boldsymbol{\delta}_t^* + \hat{v}_{t+1}) \cdot \Phi^{(3)}((\mathbf{r} - \boldsymbol{\lambda}_\mathbf{j})/\Delta\rho). \tag{16}$$

Compared with Section III-A, the proposed deformation feedforward prediction model can be completed almost without time cost (≤ 1 ms). At the same time, predicting the next path deformation can enable the robot to follow the time-varying path in a feedforward manner, improving the following accuracy.

C. Robotic RCM Navigation for Time-Varying Path

After the deformed path and its feedforward prediction were obtained, in this study, a time-varying vector field method [27], [28] with RCM constraint is introduced to guide the robot following the time-varying surgical path. Let $\mathbf{x} = \mathbf{P}_{\mathrm{tip}}$ be the current position of the instrument tip of robot B, and $\mathcal{P}(t) = \mathbf{r}(s, t)$ is the currently registered path, The closest point $\mathbf{x}^*$ on the path to the robot tip is defined as

$$\begin{cases} \mathbf{x}^*(\mathbf{x}, t) = \{\mathbf{r}(s, t) | s \in s^*(\mathbf{x}, t)\} \\ s^*(\mathbf{x}, t) = \arg\min_s \|\mathbf{x} - \mathbf{r}(s, t)\|^2 \end{cases}. \tag{17}$$

Then, the shortest distance vector between the robot tip and the path can be obtained

$$\mathbf{D}(\mathbf{x}, t) = \{\mathbf{x} - \mathbf{x}^*(\mathbf{x}, t)\}. \tag{18}$$

$D(\mathbf{x}, t) = \|\mathbf{D}(\mathbf{x}, t)\|$is a norm of the shortest distance vector $\mathbf{D}$. At the same time, the tangent vector of the current path at the closest point can be obtained

$$\mathbf{T}^*(\mathbf{x}, t) = \partial \mathbf{r}(s^*, t)/\partial s. \tag{19}$$

Given these definitions, the navigation law of the instrument tip velocity $\dot{\mathbf{x}} = \Psi(\mathbf{x}, t)$ of robot B is defined as follows:

$$\begin{aligned} \Psi(\mathbf{x}, t) &= \eta \cdot \Psi_S + \Psi_T = \eta \cdot (\Psi_G + \Psi_H) + \Psi_T \\ &= \eta \cdot \left[-G(\mathbf{x}, t) \frac{\mathbf{D}(\mathbf{x}, t)}{D(\mathbf{x}, t)} + H(\mathbf{x}, t) \mathbf{T}^*(\mathbf{x}, t) \right] \\ &\quad + \Pi_{\mathbf{T}}(\mathbf{x}, t) \frac{\partial \mathbf{D}(\mathbf{x}, t)}{\partial t}. \end{aligned} \tag{20}$$

In (20), Ψ_S is a static term and is not related to the variation of the path over time. It contains two parts: Ψ_G is a convergence term, which enables the robot to converge to the current path, and Ψ_H is the traversal item, allowing the robot to move along the path. While the last term Ψ_T is a time feedforward term, it can compensate for path deformation in advance. It represents the current deformation speed of the path at the nearest point $\mathbf{x}^*$.

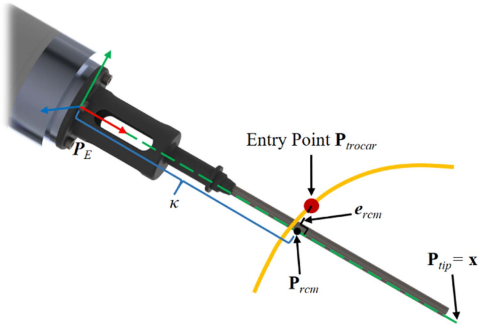


Fig. 4. Illustration of the remote center of motion constraint.

According to (16), the current speed of path deformation can be expressed as

$$\frac{\partial \mathbf{D}(\mathbf{x}, t)}{\partial t} = \hat{\mathbf{r}}(s^*, t+1) - \mathbf{r}(s^*, t). \tag{21}$$

The coefficients G and H in (20) adaptively adjust the relative weight between the convergence term and the traversal term with k_f a convergence constant. When k_f becomes larger, the weight of the convergence term increases

$$\begin{cases} G(\mathbf{x}, t) = (2/\pi) \arctan(k_f D(\mathbf{x}, t)) \\ H(\mathbf{x}, t) = \sqrt{1 - G(\mathbf{x}, t)^2} \end{cases}. \tag{22}$$

The matrix $\Pi_{\mathbf{T}}(\mathbf{x}, t)$ is the null space of $\mathbf{T}^*$ projection, which only reserves the components of the feedforward term that are in the same direction as the convergence term, given by

$$\Pi_{\mathbf{T}}(\mathbf{x}, t) = \mathbf{I} - \mathbf{T}^*(\mathbf{x}, t) \cdot \mathbf{T}^*(\mathbf{x}, t)^{\mathrm{T}} \tag{23}$$

Finally, the coefficient η is a scaling factor that adjusts the relative magnitude of the static term Ψ_S and time feedforward term Ψ_T to ensure that the overall velocity norm is constant, given by

$$\eta = -\Psi_S^{\mathrm{T}} \Psi_T + \sqrt{(\Psi_S^{\mathrm{T}} \Psi_T)^2 + v_r^2 - \|\Psi_T\|^2} \tag{24}$$

where v_r is the desired Cartesian velocity of the instrument tip.

The proof for the convergence of the navigation law can be found in [27]. The convergence condition is that the motion speed of the path is less than the speed v_r of the robot tip. This is intuitively easy to understand: a robot cannot follow a path whose movement speed is faster than its own.

It is noted that US-guided surgeries are generally performed subcutaneously. As shown in Fig. 4, the robot must control the surgical instrument moving through the epidermal entry point $\mathbf{P}_{\mathrm{trocar}}$ without lateral motion. This requires constraining the robot's motion with respect to a point known as RCM [29]. Let $\mathbf{P}_E$ be the position of the end of robot B. $\mathbf{P}_{\mathrm{rcm}}$ is the closest point to $\mathbf{P}_{\mathrm{trocar}}$ on the surgical instrument axis constructed by $\mathbf{P}_E$ and $\mathbf{P}_{\mathrm{tip}}$. According to [30], the kinematic equation of point $\mathbf{P}_{\mathrm{rcm}}$ can be written as

$$\begin{aligned} \mathbf{P}_{\mathrm{rcm}} &= \mathbf{P}_E + \kappa(\mathbf{P}_{\mathrm{tip}} - \mathbf{P}_E), \quad 0 \leq \kappa \leq 1 \\ \dot{\mathbf{P}}_{\mathrm{rcm}} &= \dot{\mathbf{P}}_E + \dot{\kappa}(\mathbf{P}_{\mathrm{tip}} - \mathbf{P}_E) + \kappa(\dot{\mathbf{P}}_{\mathrm{tip}} - \dot{\mathbf{P}}_E) \end{aligned} \tag{25}$$

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$$= \boldsymbol{J}_E\dot{\mathbf{q}} + \dot{\kappa}(\mathbf{P}_{\text{tip}} - \mathbf{P}_E) + \kappa(\boldsymbol{J}_{\text{tip}}\dot{\mathbf{q}} - \boldsymbol{J}_E\dot{\mathbf{q}})$$
$$= \begin{bmatrix} \boldsymbol{J}_E - \kappa(\boldsymbol{J}_{\text{tip}} - \boldsymbol{J}_E) \\ \mathbf{P}_{\text{tip}} - \mathbf{P}_E \end{bmatrix}^T \cdot \begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\kappa} \end{bmatrix}$$
$$= \boldsymbol{J}_{\text{rcm}} \cdot \begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\kappa} \end{bmatrix}. \quad (26)$$

The symbol $\dot{\mathbf{q}}$ represents the robotic joint speed, with $\boldsymbol{J}_{(\cdot)}$ as the current Jacobian matrix of a certain position. Based on the RCM constraint, $\mathbf{P}_{\text{rcm}}$ needs to be constantly located at the entry point, i.e., $\mathbf{P}_{\text{rcm}} \equiv \mathbf{P}_{\text{trocar}}$ and $\dot{\mathbf{P}}_{rcm} = 0$. Furthermore, the speed of the surgical instrument tip needs to satisfy the navigation law: $\dot{\mathbf{P}}_{\text{tip}} = \dot{\mathbf{x}} = \Psi(\mathbf{x}, t)$. Therefore, the extended navigation law with RCM constraint is defined as

$$\dot{\mathbf{x}}_{\text{EXT}} = \begin{bmatrix} \Psi(\mathbf{x}, t) \\ 0_{3\times 1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{J}_{\text{tip}} & 0 \\ \boldsymbol{J}_{\text{rcm}} & \end{bmatrix} \cdot \begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\kappa} \end{bmatrix} = \boldsymbol{J} \cdot \begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\kappa} \end{bmatrix}. \quad (27)$$

It can be found that the RCM error is $\boldsymbol{e}_{\text{rcm}} = \mathbf{P}_{\text{rcm}} - \mathbf{P}_{\text{trocar}}$. So the robotic kinematic control law is

$$\begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\kappa} \end{bmatrix} = \boldsymbol{J}^{\#} \cdot \begin{bmatrix} \boldsymbol{K}_{\text{tip}} & 0 \\ 0 & \boldsymbol{K}_{\text{rcm}} \end{bmatrix} \cdot \begin{bmatrix} \Psi(\mathbf{x}, t) \\ \boldsymbol{e}_{\text{rcm}} \end{bmatrix} + (\mathbf{I} - \boldsymbol{J}^{\#}\boldsymbol{J})\boldsymbol{w} \quad (28)$$

where $\boldsymbol{J}^{\#}$ is the pseudoinverse of $\boldsymbol{J}$, $\boldsymbol{K}_{\text{tip}}$, and $\boldsymbol{K}_{\text{rcm}}$ are positive definite diagonal gain matrixes, and $\boldsymbol{w} = \mathbf{0}$ is a residual input that controls the robot's other redundant degrees of freedom.

Based on the control law (28), the extended time-varying navigation method can guide the surgical instrument tip following the intraoperative deformable surgical path in a feedforward manner, compensating for tissue deformation, and making the robot's motion satisfy the RCM constraint.

D. Implementation

For the implementation of the proposed method, the robotic surgical path is represented by cubic B-splines to ensure smoothness. The resolution of the US image was resized to 256×256, and the number of control points was set to 19×19 with $\gamma = 0.02$. Hence, a total of 722 variables need to be optimized for each registration. The forgetting factor μ of FFRLS is set to 0.65. Regarding the time-varying vector field navigation algorithm, the constant k_f is set to 1.2, the desired speed norm v_r is set to 2.5 mm/s, $\boldsymbol{K}_{\text{tip}} = \mathbf{I}$, and $\boldsymbol{K}_{\text{rcm}} = 0.01 \cdot \mathbf{I}$. The proposed method was implemented in the coordinates of the US image. According to the coordinate relationship between the US image and robot B, the instrument tip speed in the US image can be converted to the speed related to robot B's base. The overall flow of the proposed method is shown in Algorithm 1.

Algorithm 1: Robotic US Autonomous Navigation.

1: **Input:** US image I_0, I_t, path $\mathcal{P}_0$, and robot tip position $\mathbf{x}$.
2: **Initialize:** Control points λ, regularization γ, forget factor μ, convergence constant k_f, and robot tip's speed norm v_r.
3: **for** $t = 1$ to T **do**
4: **Tracking Stage**
5: Calculate $\boldsymbol{\delta}^*_{t-1,t}$ of the current time step from (6)
6: Obtained current deformed path $\mathcal{P}(t)$ from (7)
7: **Prediction Stage**
8: Update the prediction model $\mathbf{y}_t$ in (10).
9: Fit α and β via FFRLS from (12)–(14).
10: Predict next path$\hat{\mathcal{P}}(t+1)$ via (15) and (16)
11: **Navigation Stage**
12: Calculate variables $\mathbf{x}^*$, $\mathbf{D}$, $\mathbf{T}^*$, G, H, and η.
13: Solve the velocity $\dot{\mathbf{x}}$ of robot tip according to (20).
14: Obtain the robotic ideal joint speed from (28).
15: **end for**
16: **Output:** ideal robotic joint speed $\dot{\mathbf{q}}$.

IV. Experiments and Results

To demonstrate the proposed method, a dual-arm robotic US surgical navigation system prototype was constructed, and a series of experiments were conducted. Experiments included a US-guided dynamic path following experiment which mainly verifies dynamic following accuracy, and an *in vitro* tissue ablation experiment which demonstrates the clinical potential.

A. Experimental Setup

The constructed robotic US navigation system prototype is shown in Fig. 5. A UR5 (Universal Robots, Odense, Denmark) robot A (six degrees of freedom, postures repeatability ± 0.03 mm) holds a US probe to conduct US imaging of a target tissue-equivalent phantom. The phantom is placed inside the laparoscopic training box and filled with silicone gel of different densities, which can be observed in the US image. A DW-PF522 ultrasound system (Dawei Company Ltd., Jiangsu, China) was applied to obtain real-time US image streams. Another UR5 robot B carries a thulium laser fiber (Raykeen Co., Ltd., Shanghai, China) through the side entry of the laparoscopic box, ablating the target area on the phantom under RCM constraint. A Laptop PC (Ryzen R5-4600H CPU and 16 G RAM) acquires the US image stream at a frequency of 30 HZ through a video capture card and calculates and predicts the phantom deformation. Simultaneously, it also controls the motion of robots A and B through the real-time data exchange client library. During the experiment, a Dynamixel XH430 motor (ROBOTIS Co., Ltd., Seoul, South Korea) drives the lead screw to move the phantom and provided its real-time reference position. The relative coordinate relationship between robot A, robot B, and the screw is calibrated by the method in [31]. After tracking and predicting the phantom deformation, according to the RCM navigation law, the PC controls robot B to follow the dynamic surgical path of the phantom. Furthermore, a video attachment is provided to demonstrate the proposed method intuitively.

The code implementation of the proposed method is completed in C++ and Matlab on the Laptop CPU platform. The deformation and interpolation of the US image and path are realized in C++ and imported as a mex function into MATLAB. The rest of the code is implemented in Matlab. Regarding the time cost of the proposed method, nonrigid registration is the most time-consuming component, taking approximately 30 ms. Other codes such as US image stream acquisition, path feedforward

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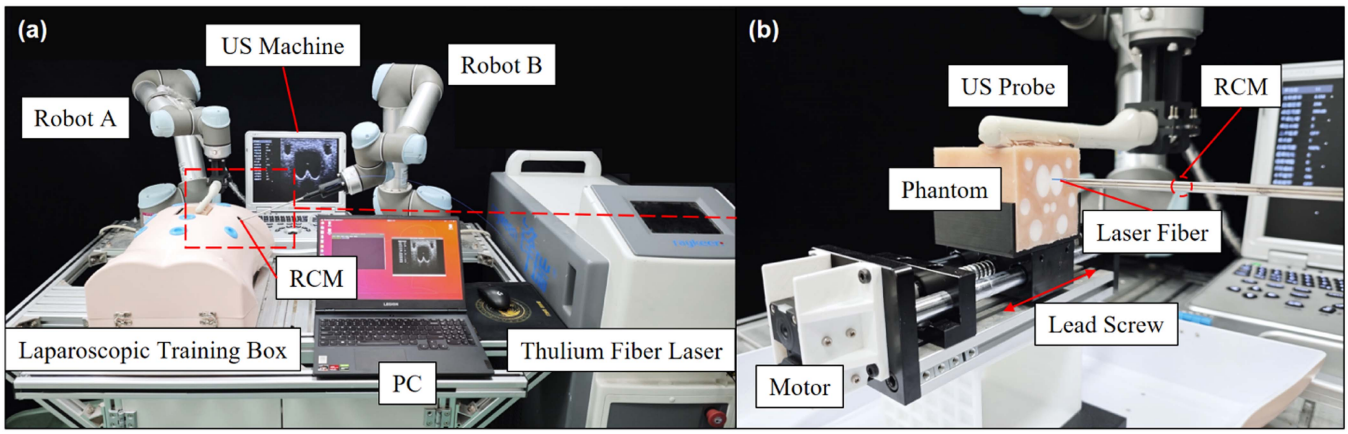


Fig. 5. (a) Setup of the constructed robotic US navigation system. (b) Perspective view of the interior of the laparoscopic training box.

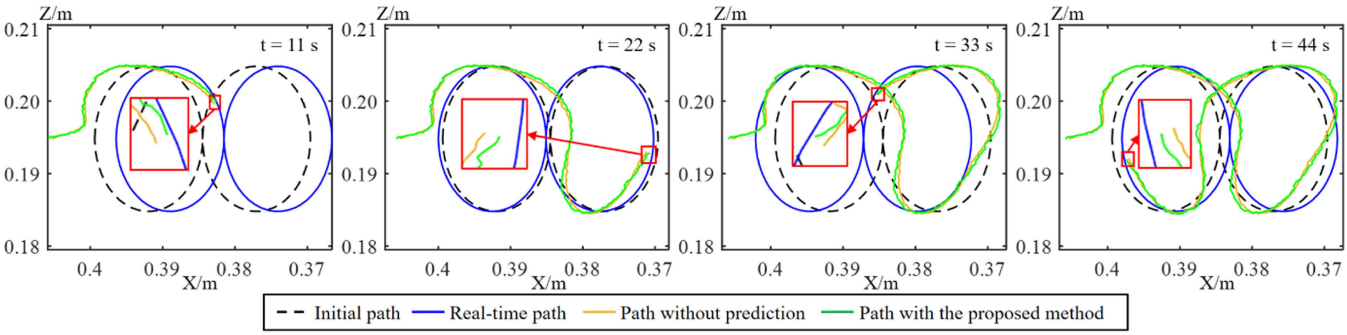


Fig. 6. Deformation of the path over time and real-time following effects of various methods.

prediction, robotic navigation, image display, etc., take about 16 ms. Therefore, the total time burden of the system is less than 50 ms, allowing the system to guide the robot's joint speed at a frequency of 20 Hz. After receiving the desired joint speed, the robotic underlying controller is responsible for servo control of the joint speed.

B. US-Guided Dynamic Path Following Experiment

In this experiment, as shown in Fig. 6, the preoperative initial path of robot B was set to an eight-like path composed of two ellipses. The screw drove the phantom to move in a sinusoidal curve on the x-axis of robot B, with an amplitude of 3 mm and a period of 15 s. The phantom moved for three periods. During this process, robot A kept the US probe still, and the US system obtained the US image stream on the moving phantom and transmitted it to the PC in real time. The PC registered the US image stream, tracked and predicted the phantom and path deformation, and then controlled robot B to follow the dynamically deformed path according to the RCM navigation law. During the following process, the positions of the robot B and the phantom were recorded for error analysis.

The navigation law in Section III-C degenerates into the method in literature [32] when without the time feedforward prediction term. Fig. 6 shows the deformation of the initial path at different moments and the robotic paths guided by these two

TABLE I
DYNAMIC PATH FOLLOWING ERRORS (MM) OF DIFFERENT METHODS

Method	Mean	Max	Std
Without Prediction	0.445	1.500	0.298
ETS	0.402	1.249	0.274
CVM	0.405	1.126	0.254
RLS	0.405	1.065	0.258
Proposed (FFRLS)	0.262	0.824	0.183

methods. It can be found that the proposed method significantly reduced the dynamic path following error compared with the method without prediction. This result is also demonstrated in Fig. 7(a), which illustrates the variation of dynamic path following errors with time. The following error of the proposed method was always within 0.83 mm, which greatly compensated for tissue deformation. Fig. 7(b) shows the RCM error during the following process, where the maximum RCM error is less than 0.3 mm, demonstrating the effectiveness of the RCM navigation law.

The proposed method was also compared with three other prediction methods: exponential smoothing (ETS) [33], constant velocity model (CVM) [34], and recursive least squares (RLS) [35]. The following errors of various methods during navigation are shown in Fig. 7(c), and corresponding statistical results are listed in Table I. ETS calculates the next predicted value based on

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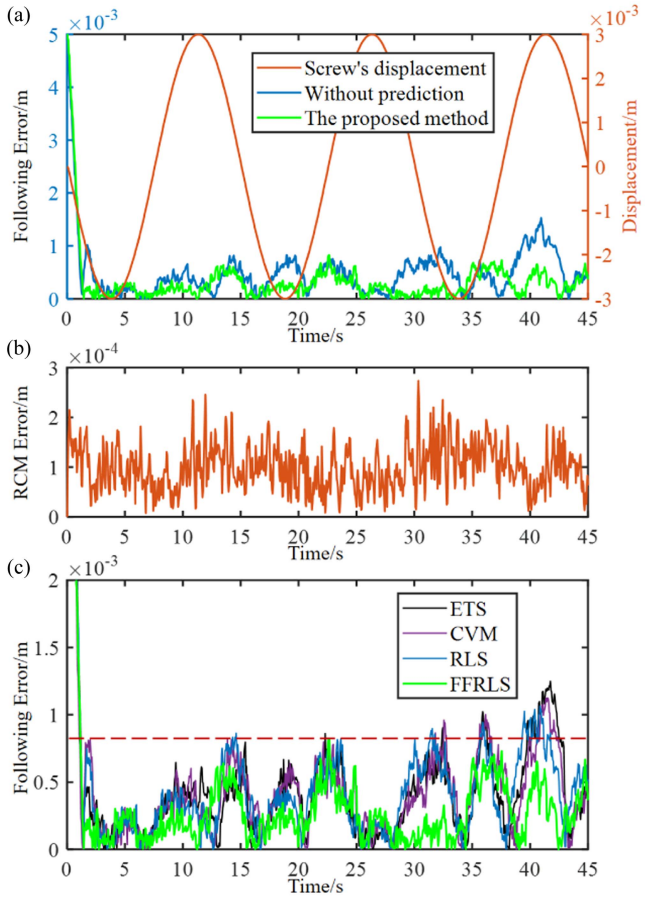


Fig. 7. (a) Value of the screw's displacement over time, and dynamic path following errors for different methods. (b) RCM error over time of robot B. (c) Following errors of different prediction methods.

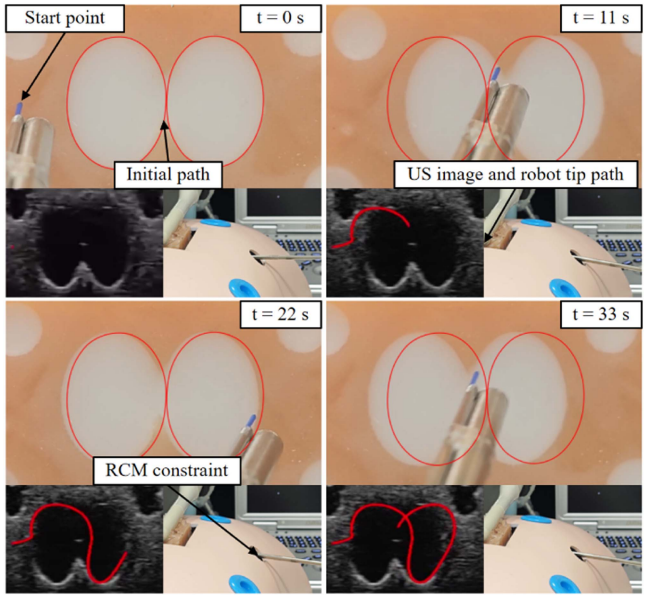


Fig. 8. Snapshots of the robotic dynamic path following the process.

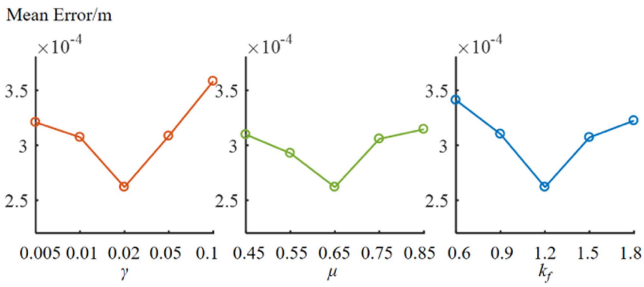


Fig. 9. Variation of the following error with various parameter values.

weighted historical data, whereas CVM assumes that the next speed is consistent with the current speed. They both have a certain time lag. With the mean and maximum following errors of the method without prediction 0.445 mm and 1.5 mm, ETS and CVM reduced the errors to 0.402, 1.249 mm, and 0.405, 1.126 mm, respectively. RLS would experience data saturation as the amount of data increases, resulting in a decrease in prediction results. It realized the mean and maximum errors of 0.405 and 1.065 mm. In general, the proposed method achieved the best results, with the mean and maximum errors being 0.262 and 0.824 mm. Fig. 8 shows the motion process of the tip of robot B guided by the proposed method. It can be found that during the deformation, the path of the instrument tip has changed greatly compared with the initially planned path, but it still stayed on the same path as the initial path relative to the phantom.

Furthermore, parameter sensitivity analysis was also conducted on the key parameters in the proposed method, including the registration regularization γ, the forgetting factor μ, and the convergence constant k_f. When one of the parameters was changed, the other parameters remained at the optimal value. The mean following results under different parameter values are illustrated in Fig. 9. It was found that when changing the values of

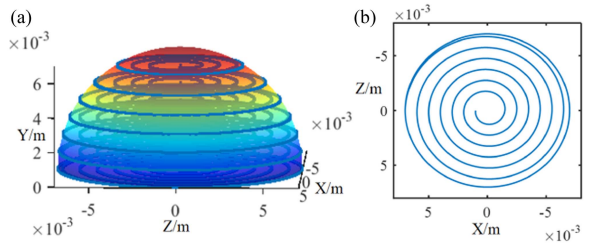


Fig. 10. (a) Hypothetical tumor hemisphere. (b) Preoperatively planned 2-D spiral path.

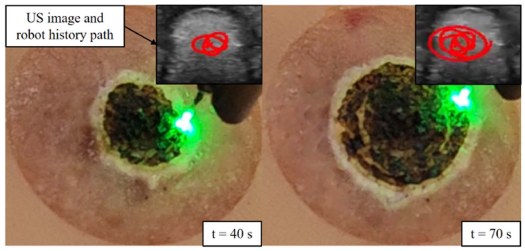


Fig. 11. Snapshots of the dynamic tissue ablation process.

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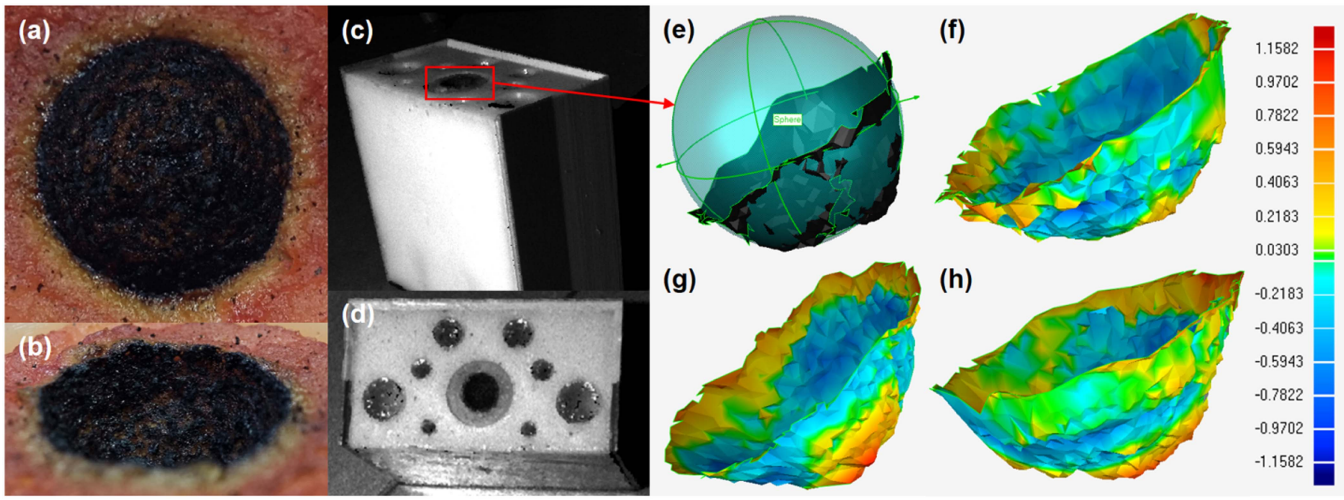


Fig. 12. (a) and (b) Appearance of the ablated area on the phantom after dynamic ablation. (c) and (d) Ablated phantom surface point set through a 3-D scanner. (e) Ablation surface is fitted to the ideal sphere. (f)–(h) 3-D error map between the surfaces and the ideal sphere.

the parameters, the overall results vary continuously, and the best following result was realized at the optimal parameters value.

C. In Vitro *Dynamic Tissue Ablation Experiment*

To further demonstrate the clinical prospects of the proposed method, an *in vitro* dynamic tissue ablation experiment was conducted. In this experiment, under the guidance of the proposed method, robot B performed laser ablation on a hypothetical tumor which was continuously moved by the screw during the ablation. The hole of the phantom was filled with pork tissue to achieve an ablation effect similar to that of human tissue. As shown in Fig. 10, the hypothetical tumor on the phantom was an ideal hemisphere with a diameter of 14 mm, and the initial planned path was composed of seven 2-D spirals with a vertical spacing of 1 mm. The amplitude of the sinusoidal motion of the phantom is set to 2.5 mm, and the period is 20 s. The ablation power of the thulium laser was set to 15 W.

The dynamic ablation experiments were performed three times. The average dynamic ablation process for the ideal hemisphere took approximately 12.5 min. As shown in Fig. 11, during dynamic deformation, the path traveled by the end of robot B was greatly different from the initially planned 2-D spiral. But on the phantom, the path maintained a good spiral shape. Fig. 12(a) and (b) shows the phantom surface after ablation, which maintained a good hemispheric appearance. To accurately acquire the ablation shape, a PhoXi 3D scanner (Photoneo Inc., Kentucky, USA) was applied to conduct a 3-D reconstruction of the ablated phantom. The reconstruction result of the 3-D scanner is shown in Fig. 12(c) and (d). After segmenting the point cloud of the ablation surface, we registered the ablation surface with the ideal sphere [see Fig. 12(e)]. Subsequently, based on the registration results, a 3-D error analysis between the ideal sphere and the ablation surface was performed. The results of the 3-D error analysis are shown in Table II. Meanwhile, the error cloud maps between the ablation surfaces and the ideal hemisphere are illustrated in Fig. 12(f)–(h). It can be found

TABLE II
STATISTICS OF DYNAMIC TUMOR ABLATION ERRORS (MM)

	Mean	Max	Std
P1	0.278	1.052	0.356
P2	0.250	1.158	0.320
P3	0.295	1.097	0.362
Total	0.274	1.102	0.346

that the average error between the ablation surfaces and the ideal spherical surface is 0.274 mm, and the maximum error is 1.158 mm. This result demonstrated that the ablation surfaces maintained good consistency with the preoperative planned ideal hemisphere, and also proved the effectiveness of the proposed navigation scheme in dynamic tissue deformation compensation under the intraoperative deformable environment.

VI. CONCLUSION

In this study, we proposed a US-guided robotic autonomous operation scheme in an intraoperative deformable environment. Through the paradigm of combining real-time tracking and prediction for tissue and path deformation, it guided the robot to automatically compensate for the intraoperative tissue deformation with RCM constraint in real time. The *in vitro* experiments on a dual-arm robotic US surgical navigation system prototype demonstrated the superiority and effectiveness of our proposed method (dynamic path following error $\leq$ 1 mm). Our future work will focus on mainly two aspects. The first is to introduce multithreading technology and graphics processing unit (GPU) hardware to further improve the real-time performance of the proposed navigation method. The second is to combine the proposed method with specific clinical scenarios, such as US-guided breast cancer biopsy and prostate ablation. We believe that the proposed navigation method can effectively reduce the adverse effects caused by tissue deformation and has the potential to promote the development of automation and precision in US-guided robotic surgery.

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